

Execution Plan for DHCP Starvation and Rogue DHCP Server Attack

Group Project

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1 Project Objective

To demonstrate the mechanics of a DHCP starvation attack followed by rogue DHCP deployment in a safely isolated lab environment. The primary goal is to understand network layer vulnerabilities and evaluate the effectiveness of Layer 2 defensive mechanisms such as DHCP Snooping and Port Security.

2 Introduction

This project demonstrates how a DHCP-based network can be exploited using a DHCP starvation attack followed by a rogue DHCP server. The idea is to first exhaust the IP pool of the legitimate server and then take control of IP assignment to redirect traffic through the attacker.

3 Core Assumptions

- **Isolated Environment:** Activities are restricted to a legally owned, isolated lab network.
- **Initial Access:** Attacker has Layer 2 access to the broadcast domain (VLAN).

- **Vulnerable Baseline:** No DHCP snooping, port security, or Dynamic ARP Inspection enabled.
- **Finite IP Pool:** DHCP server has a limited IP address pool.

4 Infrastructure Layout

- **Network Switch:** Managed switch (Cisco IOS / GNS3 / EVE-NG).
- **Legitimate DHCP Server:** Linux isc-dhcp-server.
- **Attacker Machine:** Kali Linux with Yersinia, Scapy, DHCPig, dnsmasq.
- **Victim Client:** Windows/Linux/MacOs/Android/iOS machine configured for automatic DHCP.

5 Experimental Setup

The experiment was performed in a LAN environment where all devices were connected to the same broadcast domain (single VLAN). The infrastructure consisted of an attacker machine, a legitimate DHCP server, a victim client, and a managed network switch interconnecting all devices.

Network Topology Diagram

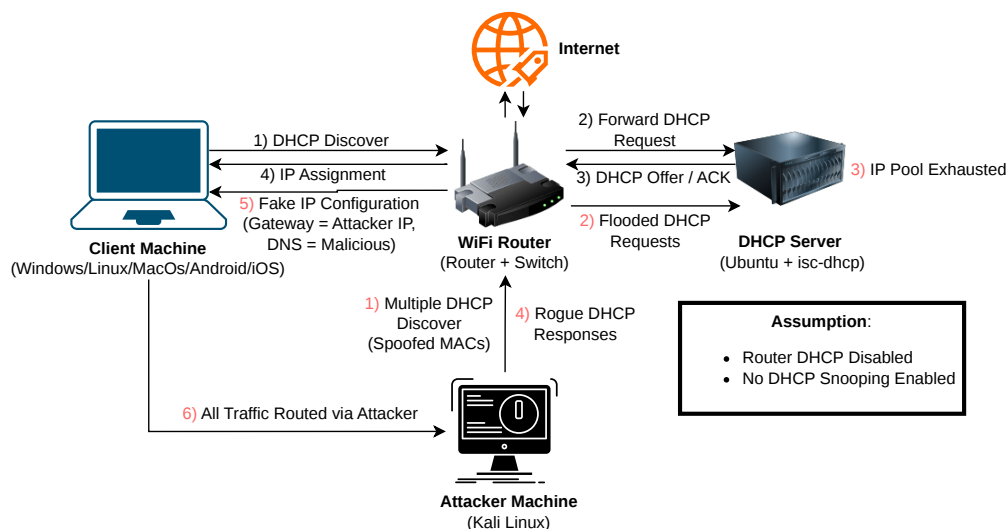


Figure 1: Network Topology for DHCP Starvation and Rogue DHCP Attack

All devices were connected through a central Layer-2 switch. The attacker machine and victim client were placed in the same broadcast domain as the legitimate DHCP server, allowing DHCP broadcast packets to be observed and manipulated.

6 Tools Used

- **Yersinia / macof / DHCPig:** Generate spoofed DHCP requests.
- **dnsmasq / isc-dhcp-server:** Setup rogue DHCP server.
- **Wireshark / tcpdump:** Packet capture and analysis.
- **Ettercap / Bettercap / Scapy:** Traffic interception and manipulation.

7 Execution Plan

Phase 1: Reconnaissance

Required Tools: Wireshark or tcpdump

Actions:

- Start packet capture.
- Apply filter: `udp port 67` or `udp port 68`.
- Identify legitimate DHCP server IP, MAC, and subnet mask.

Phase 2: DHCP Starvation

Actions:

- Use Yersinia/DHCPig to flood DHCP Discover packets.
- Spoof randomized MAC addresses.
- Monitor DHCP server lease table.

Verification: Confirm 100% IP pool exhaustion via DHCP logs.

Phase 3: Rogue DHCP Server Deployment

Configurations:

- Edit `/etc/dnsmasq.conf`.
- Set malicious Default Gateway and DNS to attacker IP.

- Assign static IP to attacker.

Execution:

- Start service: `sudo systemctl start dnsmasq`

Phase 4: Traffic Interception (MITM)

System Configuration:

- Enable IP forwarding: `sysctl -w net.ipv4.ip_forward=1`
- (Optional) Configure NAT using iptables.

Execution:

- Force victim to renew IP (`ipconfig /renew`)
- Capture redirected traffic using Wireshark.

Phase 5: Defensive Evaluation and Mitigation

Port Security Configuration:

- `switchport port-security`
- `switchport port-security maximum 2`

DHCP Snooping Configuration:

- `ip dhcp snooping`
- `ip dhcp snooping vlan 10`
- `ip dhcp snooping trust`

Verification:

- Re-run starvation tool.
- Check switch logs for violations.
- Confirm rogue DHCP packets are dropped.

8 Conclusion

This project demonstrates DHCP vulnerabilities and highlights the importance of implementing Layer 2 security mechanisms such as DHCP Snooping and Port Security to prevent network compromise.